(Please write your Enrolment No. immediately) Enrolment No. _____

MID TERM EXAMINATION

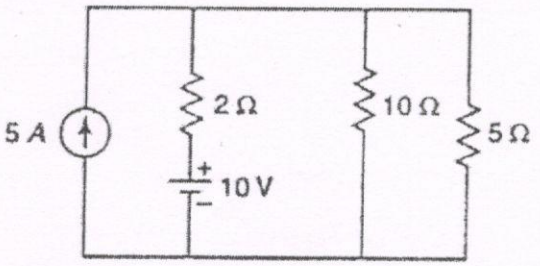
B.TECH PROGRAMMES (UNDER THE AEGIS OF USICT)

2nd Semester, April 2024

Paper Code: ES-108 **Subject: ELECTRICAL SCIENCE**

Time: 1½Hrs. **Max. Marks: 30**

Note: Attempt Q.No.1 which is compulsory and any two more questions from remaining.

Q.No.	Questions	Max. Marks	CO(s)
1(a)	What is the response of series RC circuit with AC supply. Explain with phasor diagram.	2	CO2
1(b)	If the waveform of a current has form factor 1.2 and peak factor 1.7, find the average and rms value of the current if the maximum value of the current is 100 A?	2	CO2
1(c)	Define true power and complex power?	2	CO2
1(d)	Write the expression for transforming a star connected network in a delta network.	2	CO1
1(e)	Differentiate bilateral and unilateral elements with examples.	2	CO1
2(a)	State and proof the maximum power transfer theorem having efficiency 50% with appropriate diagram and expression.	5.0	CO1
2(b)	Find the current through the 5Ω resistor using Thevenin's theorem. 	5.0	CO1
3(a)	A circuit having a resistance of 4Ω, an inductance of 0.5 H and a variable capacitance in series, is connected across a 100 V, 50 Hz supply. Calculate (i) the capacitance to give resonance (ii) the voltage across the inductance and the capacitance (iii) the Q factor of the circuit.	5.0	CO2
3(b)	Derive the expression for the resonant frequency and bandwidth of a series R-L-C circuit. Draw the phasor diagram for the same.	5.0	CO2
4(a)	Find i_1 and i_2 of the circuit shown in the following Fig. using superposition theorem.	5.0	CO1

PTO